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ON THE WEAK ERGODICITY OF NONHOMOGENEOUS CONTINUOUS-TIME MARKOV CHAINS*

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The conditions of weak ergodicity for nonhomogeneous continuous-time Markov chains are obtained in terms of intensities. The birth and death process on a finite state space is studied as a special case.

Ergodic properties of nonhomogeneous Markov chains have been studied by a number of authors. One of the main problems for such chains is the following: what properties of the intensity matrix guarantee the weak ergodicity of a chain? Some sufficient conditions were obtained by Griffeath [1] and by Zeifman [4]. More accurate estimates give more precise conditions for weak ergodicity for the finite case.

Let $X(t)$ be a nonhomogeneous continuous-time Markov chain with state space $S = 0, \ldots, N$ and intensities $a_{ij}(t)$,

$$p_i(t) = \Pr(X(t) = i), \qquad p_{ij}(s,t) = \Pr(X(t) = j \mid X(s) = i), \qquad \mathbf{p}(t) = (p_i(t)).$$

Denote by $P(s,t)$ the transition matrix for the chain $X(t)$. We assume that all intensities are locally integrable functions on $[0,\infty)$. Let Ω be the set of all stochastic vectors.

Definition 1. A Markov chain $X(t)$ is called weakly ergodic if there exists a matrix function $Q(t)$ with identical rows such that $\|P(s,t) - Q(t)\| \to 0$ as $t \to \infty$ for any $s \geq 0$.

It is noteworthy that $X(t)$ can be proved to be weakly ergodic if the above limit is equal to zero for some $s \geq 0$.

The weak ergodicity of a Markov chain and its applications were studied by Zeifman and Isaacson [5].

One can see that if $X(t)$ is a finite Markov chain, then its weak ergodicity is equivalent to the following property: $P(s,t)(\mathbf{p}^* - \mathbf{p}^{**}) \to 0$ as $t \to \infty$ for any $s \geq 0$ and any $\mathbf{p}^*, \mathbf{p}^{**} \in \Omega$.

If $X(t)$ is a homogeneous weakly ergodic Markov chain, then one can choose $Q(t) = Q$.

Definition 2. The ith state of the Markov chain $X(t)$ is called normally accessible if for any $j \neq i$ there exists a chain $i_0 = j, i_1, \ldots, i_k, i_{k+1} = i$ such that

$$\int_0^\infty a_{i_m, i_{m+1}}(u)\, du = \infty \quad \text{for any} \quad m = 0, \ldots, k.$$

Let the functions $u_1(t), \ldots, u_N(t)$ be nonnegative and locally integrable on $[0,\infty)$. Let, for any m, $\int_0^\infty u_m(t)\, dt = \infty$. Then for any s there exists t_1 such that

$$\int_s^{t_1} u_N(\tau)\, d\tau \geq 1;$$

for $t \geq t_0$

$$\int_s^t u_{N-1}(\tau) \int_s^\tau u_N(\tau_1)\, d\tau_1\, d\tau \geq \int_{t_1}^t u_{N-1}(\tau)\, d\tau;$$

hence there exists t_2 such that

$$\int_s^{t_2} u_{N-1}(\tau) \int_s^\tau u_N(\tau_1)\, d\tau_1\, d\tau \geq 1$$

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and so on. Finally, there exists t_N such that

$$\int_s^{t_N} u_1(\tau_1)\cdots \int_s^{\tau_{N-1}} u_N(\tau_N)\,d\tau_N\ldots d\tau_1 \geq 1.$$

Therefore, there exists a sequence $s_0, s_1, \ldots$ such that $s_{n+1} > s_n$ for any n and

$$\sum_{n=0}^{\infty} \int_{s_n}^{s_{n+1}} u_1(\tau_1)\cdots \int_{s_n}^{\tau_{N-1}} u_N(\tau_N)\,d\tau_N\ldots d\tau_1 = \infty.$$

Now one can see that if the chain is finite and its ith state is normally accessible, then there exists a sequence $s_0, s_1, \ldots$ such that $s_{n+1} > s_n$ for any n and for any $i_0 = j$

$$\sum_{n=0}^{\infty} \int_{s_n}^{s_{n+1}} a_{i_k,i}(u_k)\cdots \int_{s_n}^{u_1} a_{j,i_1}(u_0)\,du_0\ldots du_k = \infty. \tag{1}$$

If the chain is countable, then it is not guaranteed. For example, there is no such sequence for the chain with intensities $a_{0n} = -a_{nn} = 1/n$ and zeros in other places.

Define $M(t) = \max_i |a_{ii}(t)|$.

Definition 3. Let the ith state of the Markov chain $X(t)$ be normally accessible. It is called completely accessible if there exists a sequence s_i, satisfying (1) and such that

$$\int_{s_n}^{s_{n+1}} M(u)\,du \leq a < \infty.$$

Remark 1. In some particular cases, one can guarantee the complete accessibility of normal accessible state without any additional conditions:

(i) if a chain is finite and homogeneous;

(ii) if a chain is finite and intensities are T-periodic in t;

(iii) if a chain is finite and there exist $\alpha > 0$ and $t_0 \geq 0$ such that $a_{i_m,i_{m+1}}(t) \geq \alpha M(t)$ for any $m = 0, \ldots, k$ and almost all $t \geq t_0$.

THEOREM 1. (a) *Let $X(t)$ be a finite Markov chain.*

(i) *If it is weakly ergodic, then it has a normally accessible state.*

(ii) *If it has a completely accessible state, then it is weakly ergodic.*

(b) *Let $X(t)$ be a countable Markov chain. Then* (ii) *is correct.*

Proof. (i) Let $X(t)$ have no normally accessible state. Consider the "perturbed" Markov chain $\overline{X}(t)$ with intensities $\overline{a}_{ij}(t)$, where

$$\overline{a}_{ij}(t) = \begin{cases} a_{ij}(t), & \text{if } \int_0^{\infty} a_{ij}(u)\,du = \infty, \\ 0, & \text{if } \int_0^{\infty} a_{ij}(u)\,du < \infty. \end{cases}$$

First prove that $\overline{X}(t)$ is not weakly ergodic. Let i be an arbitrary state. Then there exists j such that any chain of intensities from j to i has zero elements. Hence, the ith state of $\overline{X}(t)$ is inaccessible from the jth state and $\Pr(\overline{X}(t) = i \mid \overline{X}(0) = j) = 0$ for any $t \geq 0$. Therefore, $\overline{X}(t)$ is not weakly ergodic. Finally, $X(t)$ is not weakly ergodic in accordance with Theorem 3.3 by Zeifman and Isaacson [5].

(ii) Assume (without loss of generality) that the state 0 is completely accessible and the state space of the chain is finite or countable. Consider the forward Kolmogorov system for $X(t)$. Let $X(s) = j$, where $s \geq 0$ and j is arbitrary. Now one has

$$\frac{dp_j}{dt} = \sum_k a_{kj}(t)p_k \geq a_{jj}(t)p_j,$$

then for $t \geq s$

$$p_j(t) \geq \exp\left(\int_s^t a_{jj}(u)\,du\right);$$

$$\frac{dp_{i_1}}{dt} = \sum_k a_{k,i_1}(t)p_k \geq a_{i_1,i_1}(t)p_{i_1} + a_{j,i_1}(t)p_j \geq a_{i_1,i_1}(t)p_{i_1} + a_{j,i_1}(t)\exp\left(\int_s^t a_{jj}(u)\,du\right),$$

then for $t \geq s$

$$p_{i_1}(t) \geq \int_s^t a_{j,i_1}(u)\exp\left(\int_s^u a_{jj}(\tau)\,d\tau + \int_u^t a_{i_1,i_1}(\tau)\,d\tau\right)du \geq \exp\left(-\int_s^t M(u)\,du\right)\int_s^t a_{j,i_1}(u)\,du,$$

then for $t \geq s$

$$p_{i_2}(t) \geq \exp\left(-\int_s^t M(u)\,du\right)\int_s^t a_{i_1,i_2}(u)\int_s^u a_{j,i_1}(\tau)\,d\tau\,du,$$

and so on.

Finally, for any j and $t \geq s$ one has

$$p_{j0}(s,t) \geq \exp\left(-\int_s^t M(u)\,du\right)\int_s^t a_{i_k,0}(u)\cdots\int_s^u a_{j,i_1}(\tau)\,d\tau\ldots du = \varphi(s,t). \tag{2}$$

Now we can apply the following known inequality: if $\mathbf{p}^*(s,s) = \mathbf{p}^*$, $\mathbf{p}^{**}(s,s) = \mathbf{p}^{**} \in \Omega$, then

$$\|\mathbf{p}^*(s,t) - \mathbf{p}^{**}(s,t)\| \leq (1 - \varphi(s,t))\|\mathbf{p}^* - \mathbf{p}^{**}\|. \tag{3}$$

Now, if $s = s_i$, $t = s_{i+1}$, $0 \leq i \leq K$, then

$$\|\mathbf{p}^*(s_0,s_{K+1}) - \mathbf{p}^{**}(s_0,s_{K+1})\| \leq \prod_{i=0}^{K}(1 - \varphi(s_i,s_{i+1}))\|\mathbf{p}^* - \mathbf{p}^{**}\|. \tag{4}$$

Therefore, any of the following conditions (i)–(iii) implies the weak ergodicity of $X(t)$:

(i) $\prod_{i=0}^{K}(1 - \varphi(s_i,s_{i+1})) \to 0$ as $K \to \infty$;

(ii) $\sum_{i=0}^{K}\log(1 - \varphi(s_i,s_{i+1})) \to -\infty$ as $K \to \infty$;

(iii) $\sum_{i=0}^{K}\varphi(s_i,s_{i+1}) \to \infty$ as $K \to \infty$.

Choose $\{s_i\}$ such that condition (1) is fulfilled. Then

$$\sum_{n=0}^{K}\varphi(s_n,s_{n+1}) \geq e^{-a}\sum_{n=0}^{K}\int_{s_n}^{s_{n+1}} a_{i_k,0}(u)\cdots\int_{s_n}^{u} a_{j,i_1}(\tau)\,d\tau\,du \to \infty \tag{5}$$

as $K \to \infty$ and estimates (4) and (5) imply the weak ergodicity of $X(t)$.

COROLLARY. *Let $X(t)$ be a finite Markov chain with proportional intensities: $a_{ij}(t) = a_{ij}\varphi(t)$. Then it is ergodic if and only if it has a normally accessible state.*

This follows from results of [3, 4] and Theorem 1.

In the case of proportional intensities and countable state space, one can obtain sufficient conditions for strong ergodicity.

Using the stability theorem by Zeifman and Isaacson [5], one can obtain sufficient conditions for strong ergodicity for a countable chain with asymptotically proportional intensities.

Remark 2. Inequalities (4) and (5) give the estimates of the rate of convergence for a weakly ergodic Markov chain.

Now let $Y(t)$ be a finite birth and death process (BDP) with intensities

$$i \longrightarrow i+1 \quad \text{(birth)} \quad -\lambda_i(t) = a_i\lambda(t);$$
$$i \longrightarrow i-1 \quad \text{(death)} \quad -\mu_i(t) = b_i\mu(t);$$

where $i = 0, \ldots, N$, $b_0 = a_N = 0$.

For example, if $Y(t)$ is queue-length process for the $M(t) \mid M(t) \mid N \mid r$ queue, then

$$\lambda_i(t) = \lambda(t); \qquad \mu_i(t) = \min(i, N)\mu(t), \qquad 0 \le i \le N + r.$$

Assume that the state 0 is accessible.

THEOREM 2. *Let all b_i $(1 \le i \le N)$ be positive.*

(i) *If the BDP $Y(t)$ is weakly ergodic, then*

$$\int_0^\infty (\lambda(t) + \mu(t))\, dt = \infty. \tag{6}$$

(ii) *If (6) is satisfied and, in addition, there exist L and $t_0 \ge 0$ such that*

$$\lambda(t) \le L\mu(t) \tag{7}$$

for almost all $t \ge t_0$, then the BDP $Y(t)$ is weakly ergodic.

Proof. (i) It follows from Theorem 1.

(ii) It follows from Theorem 1 and from condition (iii) in Remark 1. On the other hand, one can obtain a straightforward proof with the respective estimates in the following way. One has, instead of (5), for any increasing sequence s_n, $s_0 \ge t_0$,

$$\sum_{n=0}^{K} \varphi(s_n, s_{n+1}) \ge b_1 \ldots b_N \sum_{n=0}^{K} \exp\left(-R \int_{s_n}^{s_{n+1}} \mu(\tau)\, d\tau\right) \int_{s_n}^{s_{n+1}} \mu(u) \cdots \int_{s_n}^{u} \mu(\tau)\, d\tau \ldots du$$

$$= \frac{b_1 \ldots b_N}{N!} \sum_{n=0}^{K} \exp\left(-R \int_{s_n}^{s_{n+1}} \mu(\tau)\, d\tau\right) \left(\int_{s_n}^{s_{n+1}} \mu(u)\, du\right)^N, \tag{8}$$

where $R = \max(a_i L + b_i)$.

There exists a sequence s_n such that $\int_{s_n}^{s_{n+1}} \mu(u)\, du = 1$ for any n (this follows from (6) and (7)).

Then from (8) one has

$$\sum_{n=0}^{K} \varphi(s_n, s_{n+1}) \ge \frac{b_1 \ldots b_N}{N!} \sum_{n=0}^{K} \exp(-R) \tag{9}$$

and the weak ergodicity of the BDP follows from (9).

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